

BO2026 · stage-1 reproduction

Does the paper's Kalshi edge still exist in our own window, and what is it worth now?

Bartlett & O'Hara 2026 (SSRN 6615739) measured the full Kalshi history to 2026-03. This is their whole metric set re-run on our own tape, as-is, with nothing re-specified. Verdict: reproduced — every metric, with magnitudes of its own.

+31pp

single-name gap between what takers buy YES and what settles YES — the paper anchors at +28pp

+0.56¢

maker P&L per contract on single name, and it decomposes exactly as the paper says it should

2.59×

tax-to-adverse-flow ratio where the paper reports 1.52× — same structure, different number

1 of 4

groups where the VPIN cliff shows up (single name). Sports and broad-based reproduce the paper's null instead

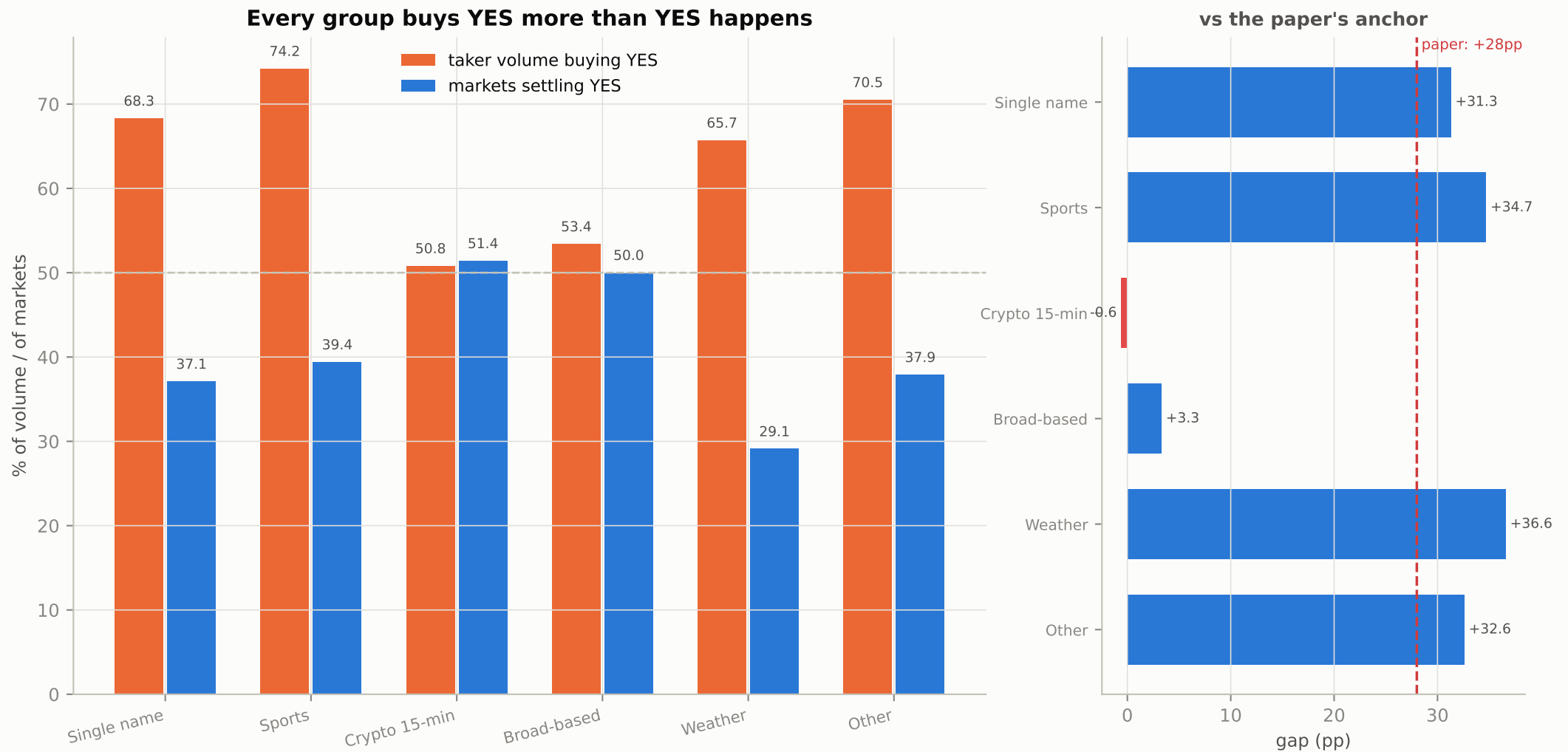
What reproduced, and what it taught us.

- ① The mechanism is intact in our window. Takers buy YES far more often than YES settles, makers earn on the other side, and the maker's edge splits into a positive frequency term and a negative magnitude term exactly as the paper's equation 7 requires. Every group reproduces the structure; none reproduces the number.
- ② The numbers do not transfer, and that became doctrine. Our tax-to-adverse ratio is 2.59× against the paper's 1.52×; our maker edge is a third of theirs on single name; their frequency edge was still rising into 2026 while ours went negative for four days mid-window. Every threshold in the paper is now treated as an initial prior only, re-estimated on a rolling window before go-live and rolled forward after.
- ③ Sports behaves like a super-single-name here, which the paper does not examine— it excludes sports from its main analysis. Taker YES 74.2% against 39.4% settling YES, the widest gap in the sample outside weather.
- ④ The VPIN cliff is real but underpowered. Direction matches on single name; the quintile buckets are too few to call it. Sports and broad-based reproduce the paper's null, which is itself a successful reproduction.

H1 · Takers buy YES; YES does not settle

YES preference

The anti-informedness gap: share of taker volume buying YES against the share of those markets that actually settled YES. The paper anchors the single-name gap at +28pp.



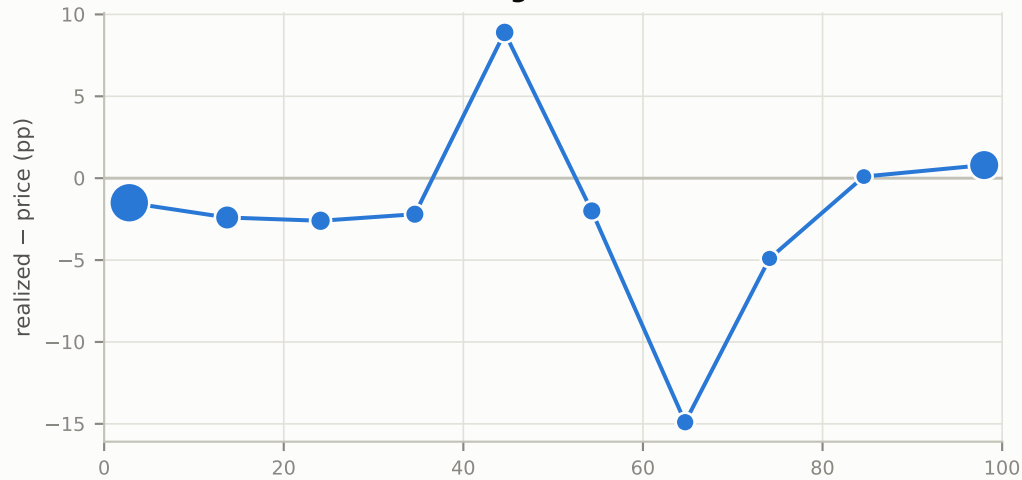
A gap is not by itself an edge — it says takers are on the wrong side of the settlement distribution, not that anyone captured it. What makes it tradable is the next page: the maker on the other side of that flow keeps a positive per-contract edge. Weather and sports post the widest gaps; the paper excludes both from its main analysis, so those two columns are ours, not a reproduction.

H2 · Calibration — where the price stops telling the truth

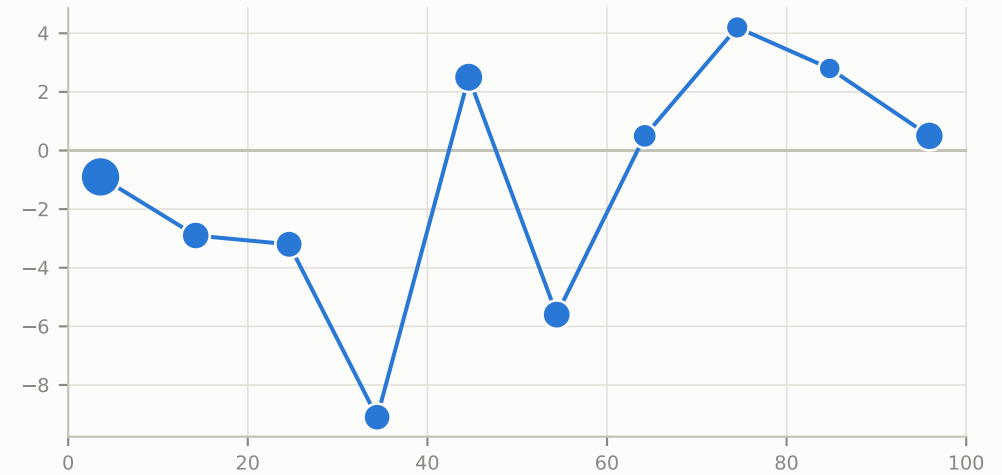
calibration

Realized YES rate minus the volume-weighted price paid, per price bucket. Zero means the price was an honest probability. The paper predicts a sag in the middle.

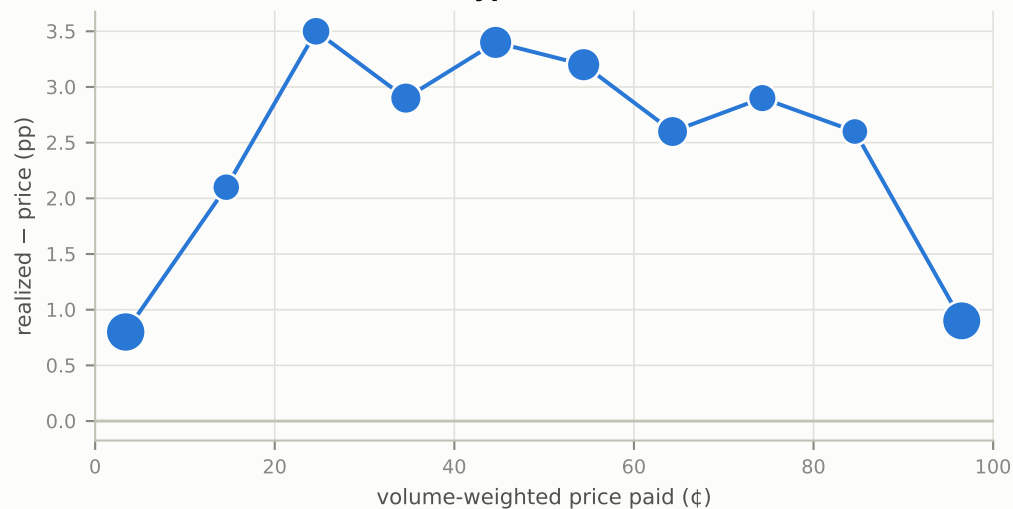
Single name



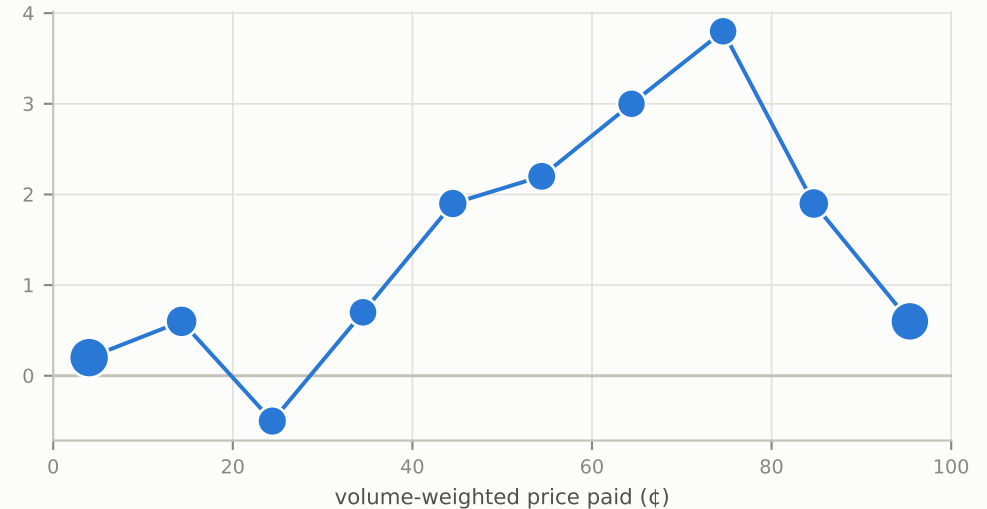
Sports



Crypto 15-min



Broad-based



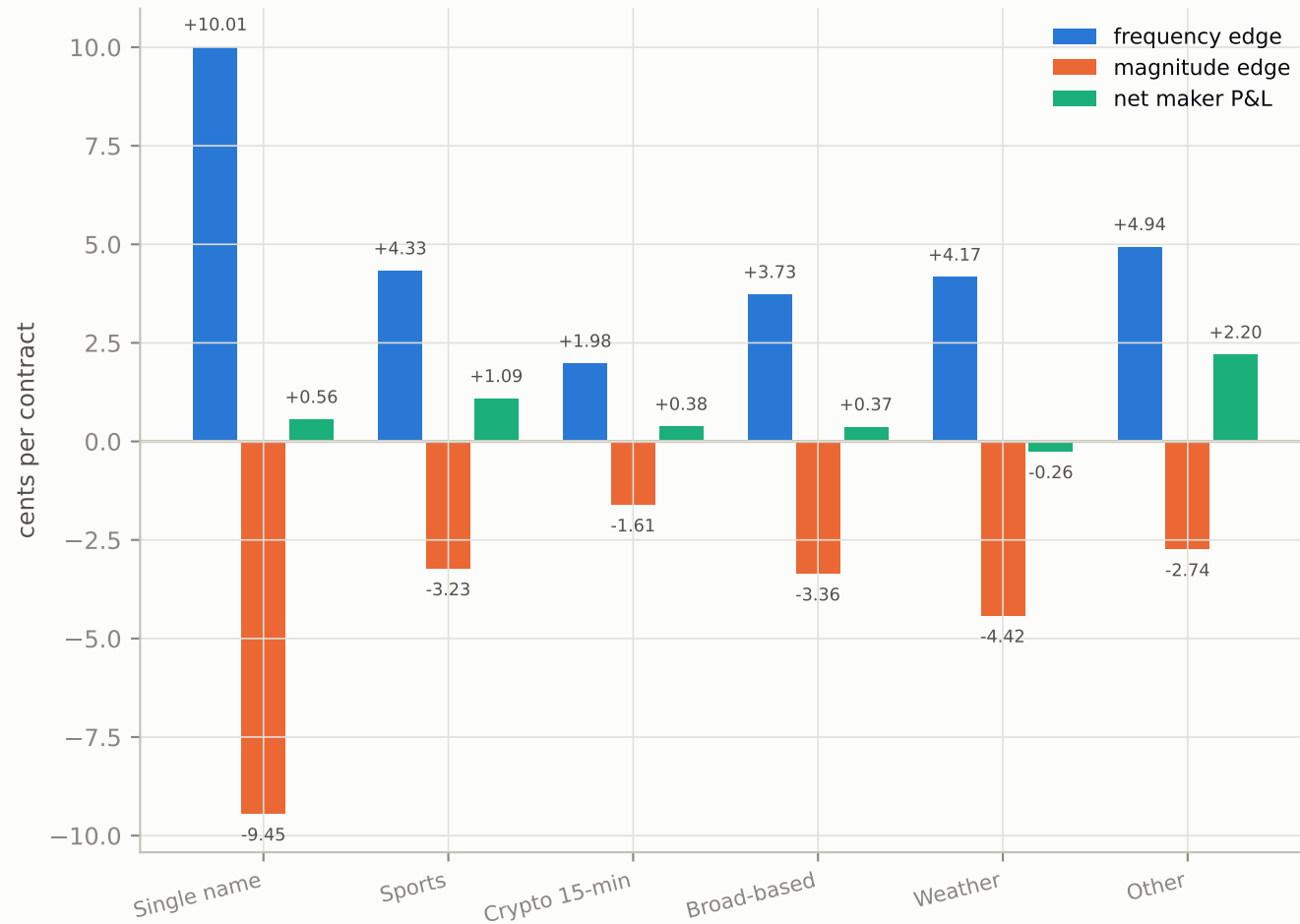
Marker area is contract volume, so the eye goes to the buckets that carry weight rather than to the noisy tails. Single name and sports sag below zero through the middle of the range — contracts there settled YES less often than their price implied, which is where the paper says the uninformed-flow premium lives. Crypto 15-min and broad-based sit above the line across the board in our window, a directional drift rather than a calibration sag.

H3 · The maker's edge splits into frequency and magnitude

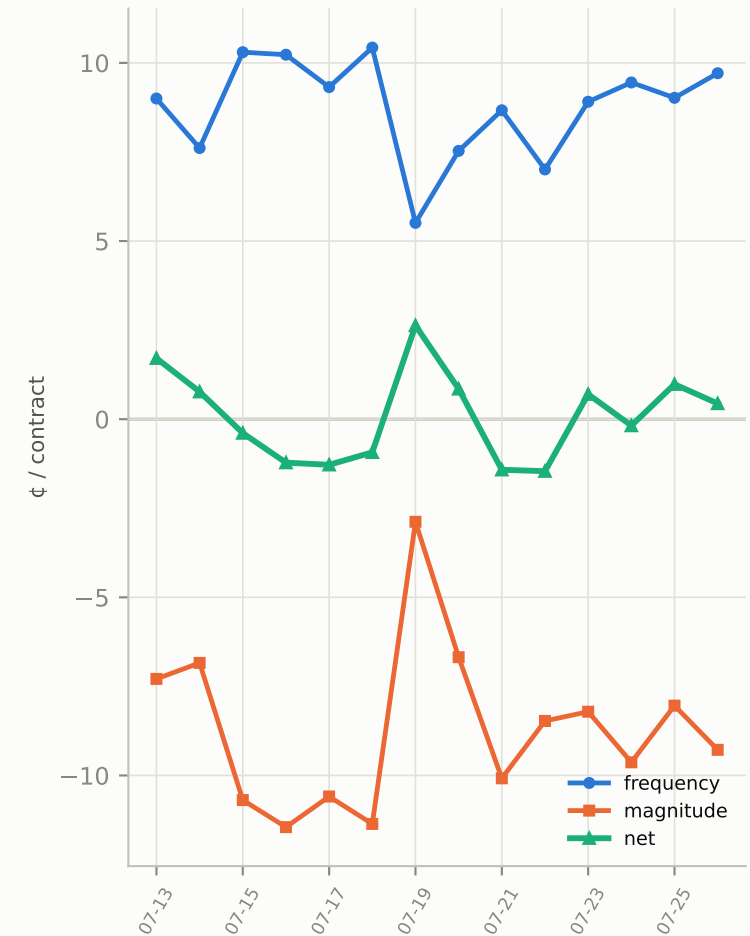
freq / magnitude

Paper equation 7: a maker facing this flow wins more often than they lose (frequency) but loses more when they lose (magnitude). The net is what is left over.

Positive frequency, negative magnitude, small positive net



single name, day by day

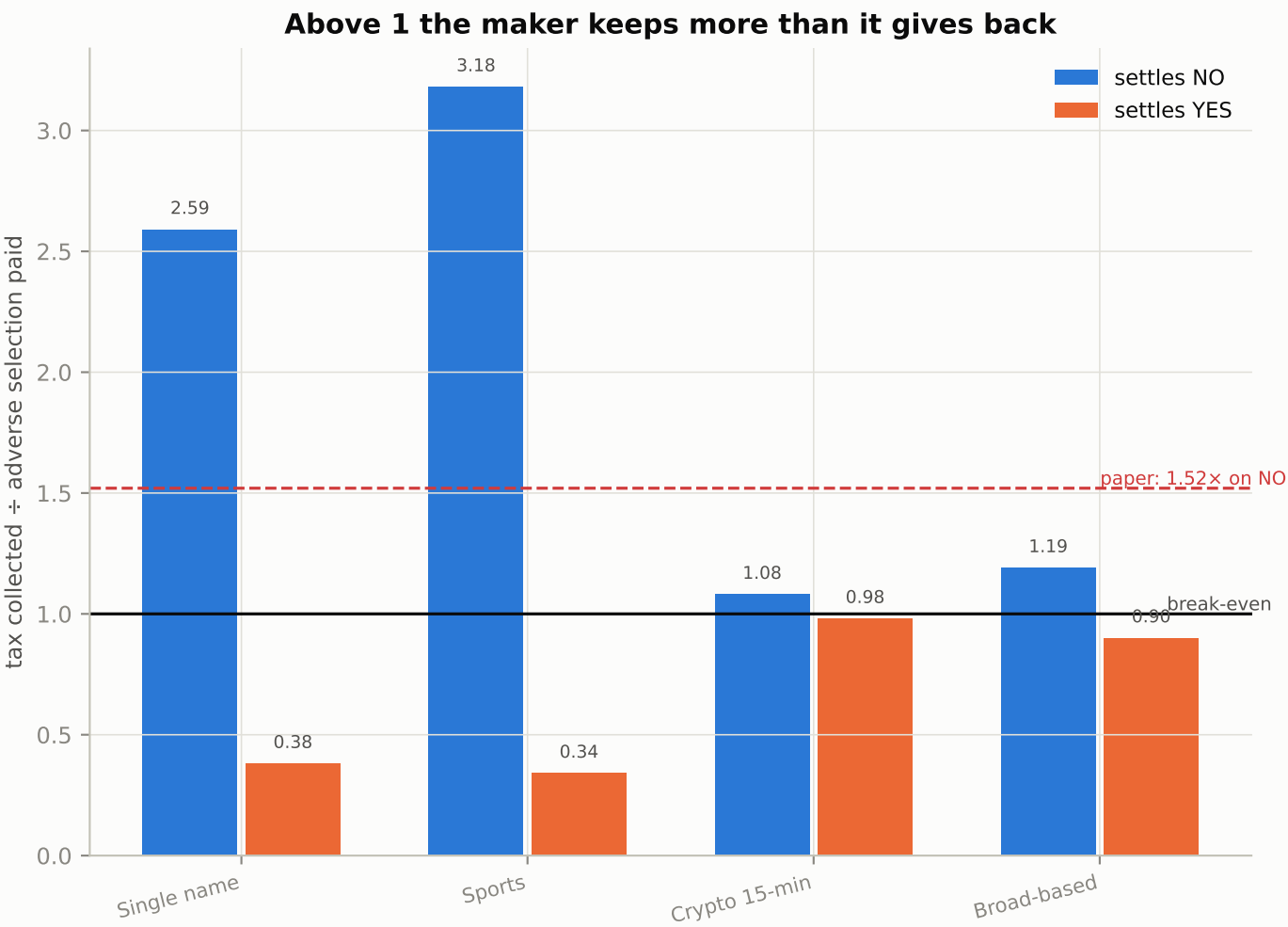


The paper reports $+1.91 = 5.31 - 3.40$ on single name; we get $+0.56 = +10.01 - 9.45$. Same decomposition, a third of the size. The day-by-day panel is why the parameter rule exists: the net term is small enough that a few bad days flip it, and it did flip negative for four consecutive days mid-window. A threshold calibrated on the paper's full-history average would have been wrong on both sides of that.

H4 · The four-cell ledger — who pays the tax

tax vs adverse

Split every settled contract by which side it settled on, then compare what the maker collected from uninformed flow (tax) against what informed flow took back (adverse). The paper predicts a ratio above 1 on NO-settling and below 1 on YES-settling.



Where the sports tax actually comes from

sports book	maker ¢/ct	taker YES%	settle YES%	volume (M)
MMA	+6.86	89.8	27.4	229
Esports	+1.34	74.9	49.3	110
Soccer	+1.30	76.7	32.9	2131
Cricket	+1.03	69.0	46.5	146
Basketball	+1.02	66.7	51.5	439
Golf	+1.01	87.4	4.5	397
Baseball	+0.69	67.6	45.4	878
Tennis	+0.34	71.6	49.9	1741
Motorsport	-2.69	82.7	25.0	12

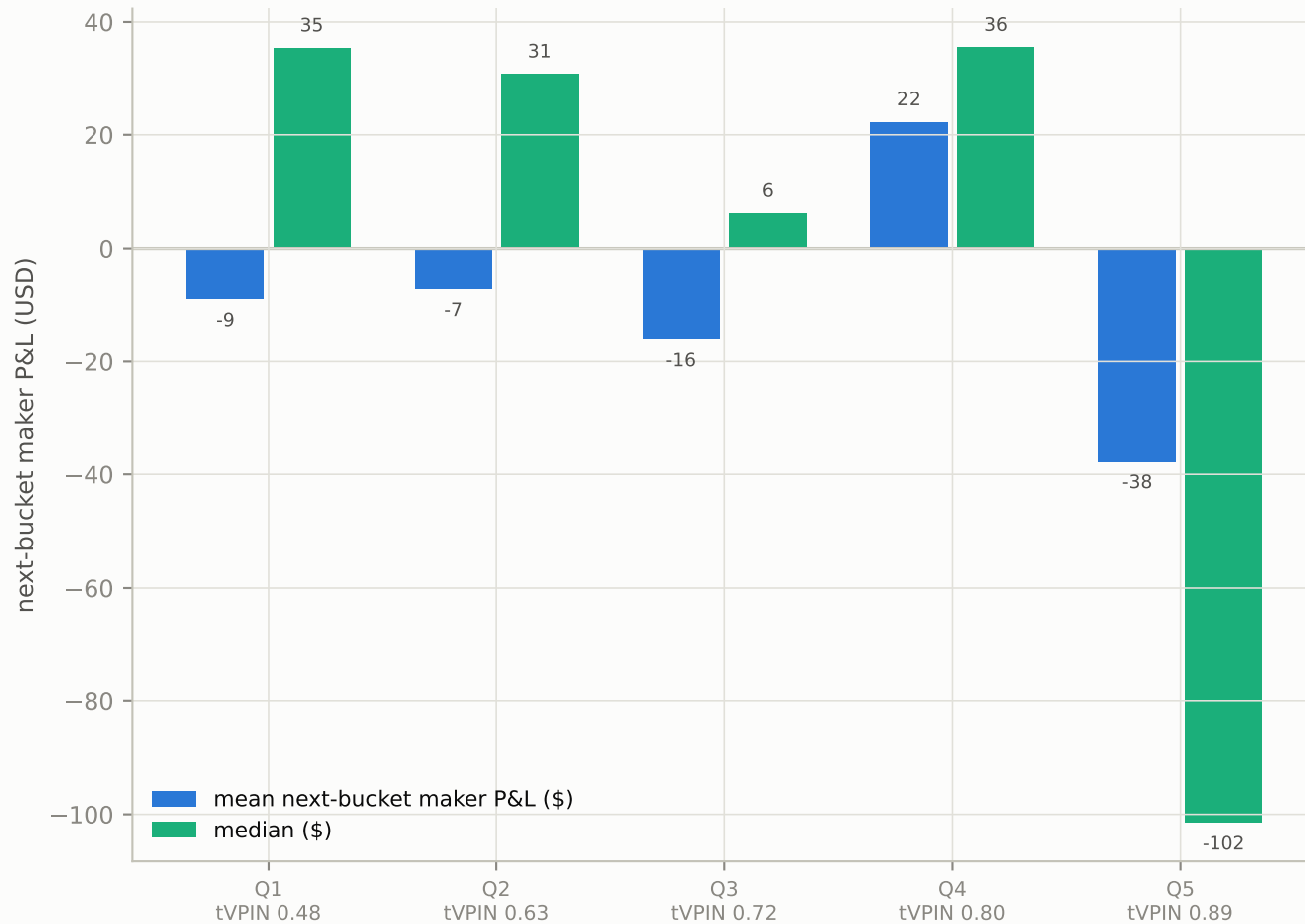
Our single-name NO-settling ratio is 2.59× against the paper's 1.52×, and the YES-settling side is 0.38× against their 0.80×— sharper on both sides, same signs. The sports table is the part the paper does not have: the tax is not spread evenly across sports, it concentrates in books where the taker YES share and the settle YES share are furthest apart (MMA: 89.8% of taker volume buys YES, 27.4% of those markets settle YES). That table is a target list for measurement, not a trade list— nothing here nets a maker fee, and the per-game sports series are exactly the ones Kalshi charges makers on.

H5 · The VPIN cliff — real, but not yet callable

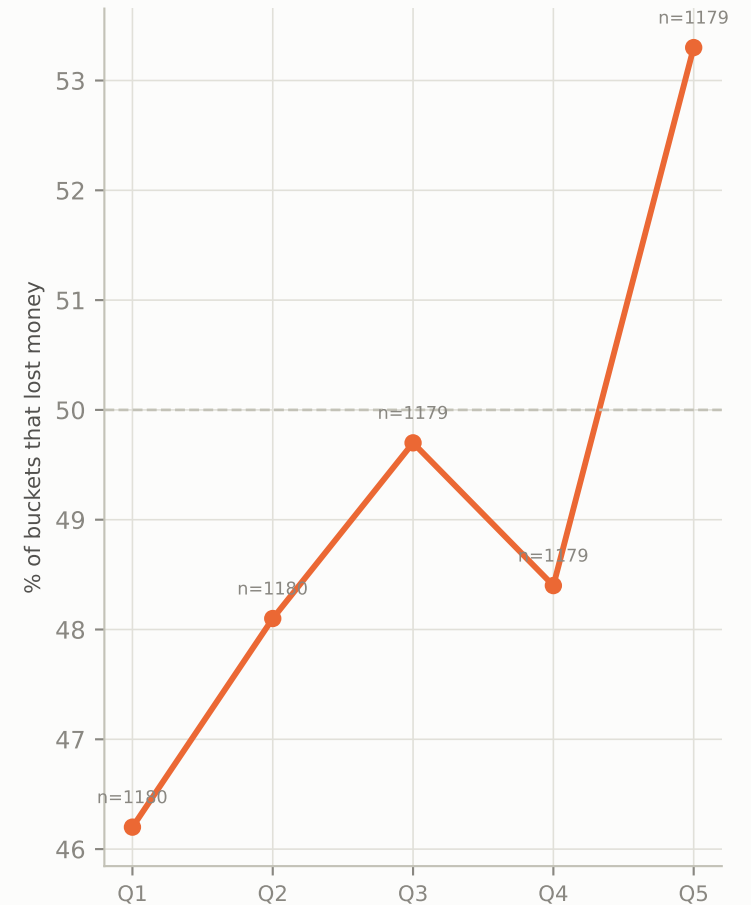
toxicity

Sort five-minute buckets by order-flow toxicity, then look at what the maker earned in the NEXT bucket. The paper predicts a cliff in the top quintile.

Single name: the top quintile is where it goes wrong



loss frequency by quintile

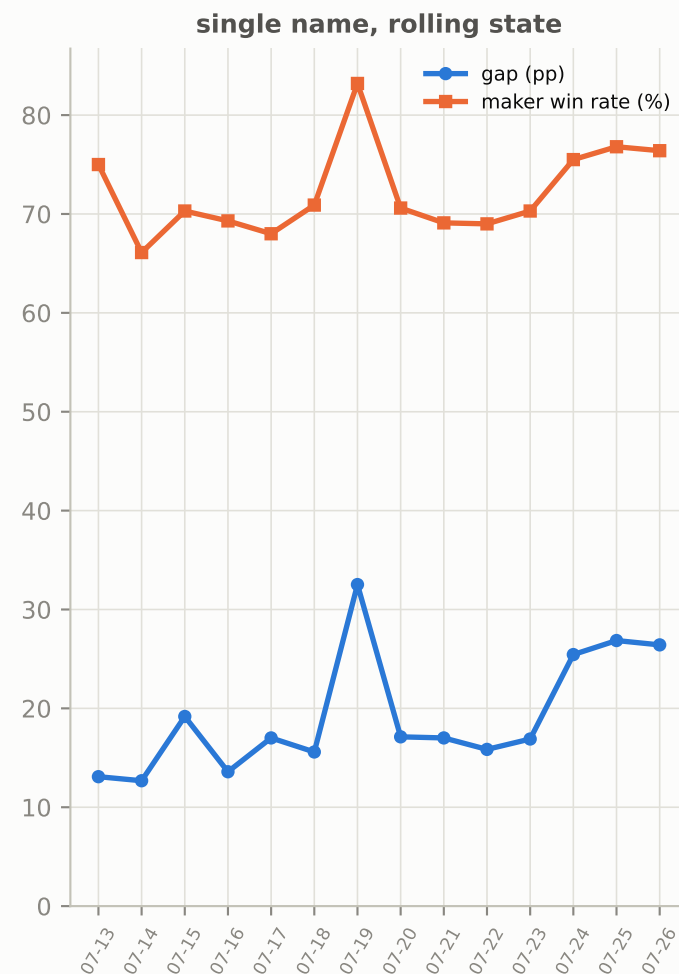
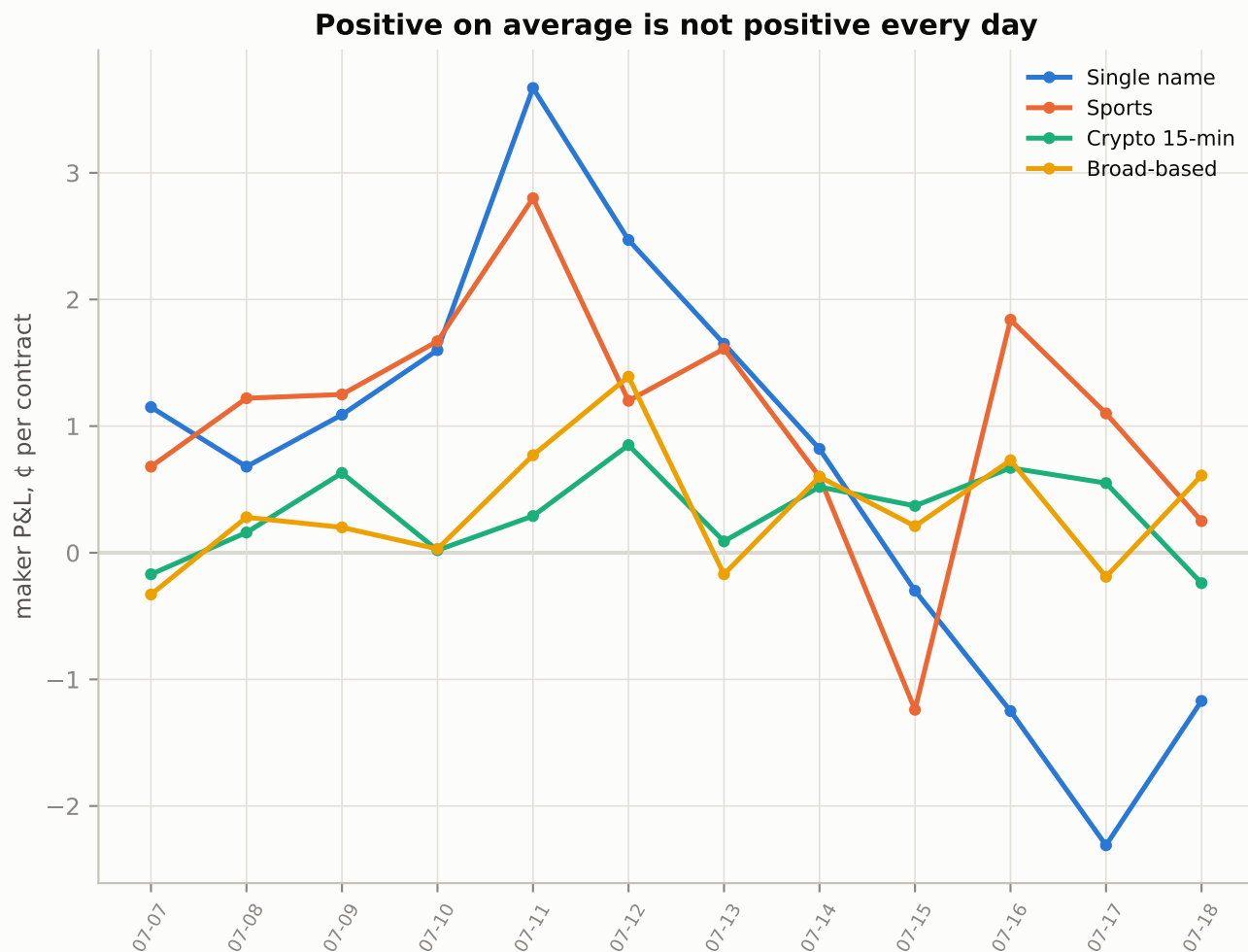


This is the one hypothesis we could not close. The direction is the paper's— the most toxic quintile has the worst next-bucket outcome and the highest loss frequency — but with roughly 1179 buckets per quintile the intervals overlap and it cannot be called. Sports and broad-based show no effect at all, which reproduces the paper's own null for those groups. The live gate currently retreats at tVPIN 0.9 and resumes at 0.8; those two numbers are priors from this chart, not conclusions, and they are re-estimated on a rolling window.

Stability — the same edge, day by day

daily

Maker P&L per contract by group and day, and the anti-informedness gap over the same window. An edge that only exists on average is not an edge we can size.



The left panel is the stage-1 window; the right panel is the live rolling state file, which keeps updating after the experiment closed. Both say the same thing: the mechanism is present on most days and absent on some. That is the entire case for treating the paper's full-history averages as priors— a strategy sized on the average would have been carrying its largest position into exactly the days where the edge was not there.

Method, and the rule this experiment produced

method

What "as-is reproduction" means here

Stage 1 re-ran the paper's metric set on our own tape with nothing re-specified: no new thresholds, no re-derived formulas, no strategy variants. Groups are behavioural proxies built from the series taxonomy and fixed before any P&L was computed, so nothing in the grouping is chosen on the outcome. Every number is gross of fees.

The parameter rule (operator, 2026-07-20 — now house doctrine)

The paper's mechanisms transfer; its numbers do not. Its sample is 2021-07 to 2026-03; we trade the market as it is now, with a different retail mix, different maker competition, and a fee schedule that keeps changing. So every numeric threshold in the paper— the VPIN cliff at 0.9, the 3¢ margin, the 1.2/1.0 tax ratios, the top-20% lambda cut, the 1.5× frequency/magnitude rule— is an initial prior only, re-estimated on our own rolling window before go-live and rolled forward after. A shadow test asks whether the mechanism still exists in our window and what it is worth now, not whether the paper's number was right. If the mechanism itself disappears— the calibration curve flattening onto the diagonal — the line is shut down, not re-tuned.

The evidence for that rule, from this very experiment

Tax-to-adverse 2.59× against the paper's 1.52×. Single-name maker edge +0.56¢ against their +1.91¢. Their frequency edge was still rising into 2026; ours went negative for four consecutive days inside a twelve-day window. Three different ways of being right about the mechanism and wrong about the magnitude.

Line separation

Paper-as-is results are BO-x lines; our own variants are RC-x and never mixed into the same P&L attribution. That separation exists so a variant that works cannot be credited to the paper, and a paper result that stops working cannot be rescued by a variant.

group	prints	markets	contracts (M)	days
Single name	906,614	2,275	52	12
Sports	40,876,376	68,371	6089	12
Crypto 15-min	25,737,715	8,855	1521	12
Broad-based	5,693,677	36,879	536	12
Weather	1,236,554	10,665	30	12
Other	4,149,966	13,088	514	12